Product Suggestions and Clusters

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Subject: Product suggestions and customer groups for Q4 planning

What we built

We built two things you asked for: a “Customers who bought this also bought…” box for every product page, and a map of our main customer groups for marketing. We used two years of real orders (December 2009 to December 2011), and we tested everything on the last two months, which the model never saw.

Which method we chose and why

We used a method called Truncated SVD. It finds about 100 hidden “themes” in what customers buy (for example, Christmas lights or heart decorations) and then describes every product and every customer by those themes. We chose it because our order table is almost empty (each customer buys only a small part of the catalog), and SVD works directly on that kind of data, so it stays small and fast.

What we lose by compressing

"We keep the main buying patterns, which account for about 45% of the information, and drop one-off purchases and smaller patterns that do not repeat across many customers. So, the model remembers that a shop likes heart decorations, but it may forget a single purchase that does not fit its usual taste (for example, one lunch bag bought by a decoration shop).

How big and how fast

The full table would take 161.4 MB, and the compressed result takes 7.3 MB (about 22 times smaller). Building the model takes under 0.2 seconds, and making suggestions for all 1,930 test customers takes 0.3 seconds, so the nightly job is no longer a problem.

Are the suggestions better than best sellers?

Yes, after one improvement. Showing every customer, our 10 best sellers gave at least one correct product for 26.6% of customers. Our model alone reached 23.4%, so we added a small bonus for popular products, and the combined version reached 33.1%. So if, for example, 100 shops visit the site, about 33 of them will see something they really go on to buy, against about 27 with best sellers.

Customer groups for marketing

We found five groups, named by the occasion they buy for: Year-Round Gifting (1,920 customers), Home Styling (1,275), Rainy-Day Play (1,041), Children’s Parties (697) and Carry & Store (540). Children’s Parties is our most international group (only 68% UK), so it suits a campaign for European shops.

What could go wrong

We chose the popularity bonus based on the test results, so 33.1% is probably a little too good. New customers and new products get no suggestions until they have some orders, and products bought by fewer than 10 customers are never suggested. Our test period (October to December) is the Christmas season, so results in other months may differ.

Next steps

We suggest testing the box on a small part of the website first, checking the popularity bonus on newer data, and rebuilding the model every night with the latest orders.